

Bahram Hedayati

+4915129728520 | hedayati@ccs-labs.org | [LinkedIn](#) | [Google Scholar](#) | Berlin, Germany

EDUCATION

- **Polytechnic University of Milan** Sep 2022 - Jul 2025
Master in Telecommunication Engineering - Communication Networks and Internet track Milan, Italy
 - Thesis Title: Entropy-driven Effective Tumor Detection using Nanoscale Medical Agents
 - Final Grade: 97/110
- **Payame Nour University** Feb 2012 - Feb 2014
Course-based Master in Computer Engineering - Software track Tehran, Iran
 - Final Grade: 16.51/20.00
- **University of Science and Art** Sep 2009 - Sep 2011
Bachelor in Software Engineering Yazd, Iran
 - Final Grade: 16.01/20.00
- **Azad University** Sep 2007 - Aug 2009
Associate in Computer Fasa, Iran
 - Final Grade: 18.00/20.00

MASTER'S THESIS SUMMARY

Title: Entropy-driven Effective Tumor Detection using Nanoscale Medical Agents [Thesis]
Developed an agent-based simulation to model a tumor microenvironment mathematically. Integrated an information-theoretic framework for physiological signal analysis using molecular communication to propose an effective and efficient navigation algorithm for drug carriers, aiming at tumor detection. The utilized principles are highly transferable to other areas of scientific and engineering problems.

Thesis Advisor: Professor Maurizio Magarini

Thesis Evaluation: Awarded 7.0 out of 7.0 points by the examination committee

INTERNATIONAL PUBLICATIONS

- [1] Bahram Hedayati, Karthik Reddy Gorla, Yue Sun, Yifan Chen, Maurizio Magarini. **Bio-Inspired Navigation for Intelligent Nanoscale Drug Delivery Systems based on Chemotaxis and Entropy**, Manuscript submitted to *IEEE Transactions on Molecular, Biological, and Multi-Scale Communications*, 2026.
- [2] Bahram Hedayati, Karthik Reddy Gorla, Yue Sun, Yifan Chen, Maurizio Magarini. **Preliminary Implementation of Entropy-driven Tumor Detection within a Blood Vessel Network**, Extended abstract published in *The 10th Workshop on Molecular Communications*, April 2026, Istanbul, Turkey. [Full Paper]
- [3] Bahram Hedayati, Karthik Reddy Gorla, Maurizio Magarini. **Entropy-driven Effective Tumor Detection using Nanoscale Medical Agents**. Published in *The 12th ACM International Conference on Nanoscale Computing and Communication*, October 2025, Chengdu, China. [Full Paper]
- [4] Bahram Hedayati, Karthik Reddy Gorla, Maurizio Magarini. **Semantic Information-Driven Targeted Drug Delivery for Cancer using Molecular Communication**. Published in *The 20th conference on Computational Intelligence methods for Bioinformatics and Biostatistics*, September 2025, Milan, Italy. [Abstract] [Poster]

WORK EXPERIENCE

- **Technical University of Berlin, Department of Telecommunication Networks (TKN)** May 2026 - Now
Research Assistant Berlin, Germany
- **Politecnico di Milano, Department of Electronics, Information and Bioengineering (DEIB)** Jan 2026 - Feb 2026
Research Assistant Milan, Italy
- **Amin Computer Repair Center** Jul 2016 - Jul 2022
Software Expert Fasa, Iran
- **Azad University** Feb 2015 - Sep 2017
Lecturer (Part-time) Fasa, Iran
- **Amin Computer Repair Center** Oct 2011 - Dec 2014
Software Expert Fasa, Iran

CERTIFICATES

• Artificial Intelligence Career Competition	Feb 2025
• Data Scientist Academy	Feb 2024
• Advanced Recommender Systems	Jan 2024
• Generative AI Primer	Nov 2023
• Introduction to Data Science in Python	Jun 2022
• Cluster Analysis in Data Mining	Jun 2022

SKILLS

- **Research Proficiencies:** Experimental Design and Data Analysis, Simulation of Dynamic Processes, Literature Review and Critical Evaluation, Research Presentation and Scientific Writing, LaTeX for Technical Reports
- **Theoretical Knowledge:** Statistical Inference, Probability Theory, Linear Algebra, Information Theory, Signal Processing, Game Theory
- **Biomedical Signal Processing:** Time-domain and frequency-domain analysis of physiological signals, Feature extraction, Noise reduction and artifact removal, Spectral Analysis
- **Modeling Skills & Tools:** COMSOL Multiphysics, Mathematical Modeling, Agent-based Modeling, PDE-based Simulations, Stochastic Simulations (Monte Carlo, Markov Chains, Survival Analysis)
- **Machine Learning and AI:** Pattern recognition, Feature extraction, Clustering, Time Series Analysis, Classification
- **Programming Languages & Frameworks:** C, C++, C#.NET, Python, Qt, SQL, MQL, MATLAB
- **IDEs:** PyCharm, Visual Studio, Google Colab, Jupyter Notebook, Qt Creator
- **Version Control:** Git, GitHub
- **Linux Distributions:** Fedora, Kali-Linux, Ubuntu
- **Network Tools:** Wireshark, Net2Lab, Cisco Packet Tracer, OMNET++

ACADEMIC PROJECTS

• Electrocardiography (ECG) Signal Analysis <i>Tools: Signal Processing, Python, NumPy, Pandas, Scikit-learn, Matplotlib, Data Processing</i>	Oct 2024 - Feb 2025
• Signal Processing of a Plant's Bio-electrical Activity <i>Tools: Action Potential Analysis, Filtering, Python, SVM, KNN, Regression, Decision Trees</i>	Mar 2024 - Jul 2024
• Traffic Forecasting of the GENAT Research Network <i>Tools: Python, NetworkX, K-Neighbor Regressor, LSTM</i>	Apr 2024 - Jun 2024
• Distance Analysis of current location to different servers using Round Trip Time (RTT) <i>Tools: Python, Web Scraping, BeautifulSoup, regression</i>	Mar 2024 - Apr 2024
• Analysis of a queuing system using priority scheduling with general services <i>Tools: C++, OMNET++</i>	Dec 2023 - Feb 2024
• Analyzing Traffic Dynamics of a Mobile Radio Network (MRN) <i>Tools: Python, Folium, time series analysis, z-score normalization, and K-Means</i>	Apr 2023 - May 2023

HONORS

• Merit-based DSU (Diritto allo Studio Universitario) Tuition Waiver and Scholarship <i>Polytechnic University of Milan, Italy</i>	Sep 2022 - Feb 2025
• Merit-based Tuition Reduction <i>Payame Nour University, Tehran, Iran</i>	Feb 2012 - Feb 2014
• Merit-based Tuition Reduction <i>University of Science and Art, Yazd, Iran</i>	Sep 2009 - Sep 2011

VOLUNTEER / ACADEMIC SERVICE

• Student Volunteer at IEEE Wireless Communications and Networking Conference	Mar 24-27 2025
• Student Volunteer at ACM Conference on Nanoscale Computing and Communication	Oct 28-30, 2024

LANGUAGES AND INTERESTS

Languages: English (Advanced), Persian (Native)
Interests: Playing Chess, Running, Photography